Elastic Load Balancer & Auto Scaling in AWS Lesson

Skills you will learn

Cloud Computing: Monitoring & Scaling

Lesson Overview

In this lesson, you will learn about **Elastic Load Balancer (ELB)** and **Amazon EC2 Auto Scaling**, two key AWS services that help ensure **high availability, automatic traffic distribution, and resource scalability**. By the end of this lesson, you will understand how these services improve application resilience and performance.

Learning Objectives

By the end of this lesson, you will be able to:

1. Understand the role of ELB and EC2 Auto Scaling in AWS.
2. Differentiate between the different types of AWS load balancers.
3. Explore the components and functionalities of EC2 Auto Scaling.

1. Introduction to Elastic Load Balancer (ELB)

AWS **Elastic Load Balancer (ELB)** is a service that **automatically distributes incoming traffic** across multiple targets such as EC2 instances, containers, and IP addresses. It ensures that applications remain highly available and scalable.

Key Features of ELB

✅ **High Availability:** Traffic is distributed across multiple **Availability Zones** for fault tolerance. ✅ **Layer 4 & Layer 7 Load Balancing:** Supports **TCP, UDP, HTTP, and HTTPS** protocols. ✅ **Security:** Works with **AWS VPC security groups** and **AWS Shield** to protect applications. ✅ **Health Checks:** Detects unhealthy instances and automatically reroutes traffic to healthy ones.

Types of Elastic Load Balancers

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AWS provides different types of load balancers, each optimized for different use cases:

Load Balancer Type	Functionality	Ideal Use Case
Application Load Balancer (ALB)	Routes HTTP/HTTPS traffic at Layer 7, supports host-based and path-based routing	Web applications, microservices, APIs
Network Load Balancer (NLB)	Handles TCP/UDP traffic at Layer 4, optimized for high performance	Real-time gaming, financial transactions
Gateway Load Balancer (GLB)	Directs traffic to virtual appliances and scales them automatically	Network security appliances, traffic inspection

2. Amazon EC2 Auto Scaling

Amazon **EC2 Auto Scaling** enables automatic scaling of compute resources **based on demand**. It ensures that the right number of instances are running to meet your workload while **minimizing costs**.

How EC2 Auto Scaling Works

✅ **Automatically launches or terminates instances** based on configured policies. ✅ **Integrates with ELB** to distribute traffic efficiently. ✅ **Maintains high availability** by replacing unhealthy instances automatically. ✅ **Optimizes cost** by scaling up/down based on demand.

Components of EC2 Auto Scaling

Component	Description
Launch Template	Defines instance configurations like **AMI, instance type, security groups, and user data**.
Auto Scaling Group (ASG)	A group of EC2 instances managed together. Defines **minimum, maximum, and desired capacity**.
Scaling Policy	Defines how instances should be added or removed (e.g., **CPU-based scaling, scheduled scaling**).

3. Scaling Policies for Auto Scaling Groups

Amazon EC2 Auto Scaling allows flexible scaling based on different policies:

Scaling Policy	Functionality
Manual Scaling	Manually adjusts instance count based on expected traffic.

Scaling Policy	Functionality
Dynamic Scaling (Target Tracking)	Adjusts capacity based on **CloudWatch metrics** (e.g., CPU utilization).
Step Scaling	Adds/removes instances in steps based on thresholds.
Scheduled Scaling	Scales based on a pre-defined schedule (e.g., increase capacity at peak hours).

Example: Dynamic Scaling Based on CPU Utilization

1. Create an Auto Scaling policy to launch an **EC2 instance** when CPU utilization exceeds **70%**.
2. Define a **CloudWatch alarm** to monitor CPU usage.
3. If the threshold is exceeded, **Auto Scaling launches new instances** to balance the load.

4. Summary & Key Takeaways

- Elastic Load Balancer (ELB) ensures fault tolerance and distributes traffic efficiently.
- Three types of ELBs (ALB, NLB, GLB) cater to different traffic needs.
- EC2 Auto Scaling automatically adjusts compute capacity to meet demand.
- Scaling policies enable manual, dynamic, and scheduled scaling.

Next Steps & Hands-On Practice

- Try setting up an Elastic Load Balancer and an Auto Scaling Group using the AWS Management Console.
- Experiment with different scaling policies and monitor traffic distribution.

🚀 **Keep exploring AWS for optimized performance and cost efficiency!** 🎯

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